# Cluster Size vs $E_T$ — Per-Period, Per-Selection

Data vs MC with truth-vertex reweighting and DI mixing

PPG12

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**What is plotted.** Per-cluster mean tower count $n_{\text{owned}}$ (from `cluster_ownership_array`) and 7×7-window bounding-box widths $\Delta i_\eta$, $\Delta i_\phi$ vs reco $E_T$, split per crossing angle, for three orthogonal cluster selections:

- **baseline**: fiducial $|\eta_{\text{reco}}| < 0.7$ and reco vertex $|z_r| < 60\,\text{cm}$ only.
- **common cut**: baseline plus the PPG12 common-region cut block from `config_showershape_{1p5rad,0rad}` (`cluster_prob`, $e_{11}/e_{33} < 0.98$, $\langle w_{\eta,\text{cogx}}\rangle < 2$, NPB score $> 0.5$).
- **tight cut**: common plus the parametric tight block (ET-dependent width and energy-fraction bounds and the `base_v1E` BDT threshold bdt $> -0.02\,E_T + 0.86$).

Sim weighting uses the data-driven iterative truth-vertex reweight $w(z_h)$ from `efficiencytool/truth_vertex_rewe` (per-period `reweight.root`; method described in `truth_vertex_reco_check.pdf`) consumed the same way `ShowerShapeCheck.C` does. Data is filtered by run range (51274–54000 for 1.5 mrad, 47289–51274 for 0 mrad). DI mixing uses $\langle y\rangle_{\text{mix}} = (1-f)\langle y\rangle_{\text{single}} + f\langle y\rangle_{\text{double}}$ applied bin-by-bin to the pooled-MC means, with $f = 0.079$ (1.5 mrad) or 0.224 (0 mrad).

**Key findings.** All numerical values cited below are measured from the pooled TProfiles in the two-bin average $E_T \in [19, 21]$ GeV of the split container, after the truth-vertex reweight. See Appendix A for the full table.

- **At baseline, truth-vertex-reweighted jet MC already tracks data closely.** At $E_T = 20$ GeV, 1.5 mrad: $\langle n_{\text{owned}}\rangle_{\text{data}} = 15.86$ vs $\langle n_{\text{owned}}\rangle_{\text{jet MC}} = 15.38$, a **+0.5-tower gap** (vs +2.3 reported in March without per-period reweighting). Photon MC baseline is 10.44 — the familiar signal/background separation.
- **The common cut removes the noise-like tower excess, preferentially from data.** At 1.5 mrad and $E_T = 20$ GeV, $\langle n_{\text{owned}}\rangle$ drops from $15.86 \to 11.01$ in data (a 31% decrease), $15.38 \to 14.33$ in jet MC (only 7%), and $10.44 \to 10.39$ in photon MC (essentially unchanged). The common cut's NPB + $w_\eta$-bound block targets noise/threshold contamination which is far more prevalent in data than in MC.
- **The tight cut converges data, jet MC, and photon MC to the signal cluster size.** At $E_T = 20$ GeV, 1.5 mrad: data tight 9.92, jet MC tight 10.94, photon MC tight 10.21 — a ±5% spread around a common ∼10.4 tower signal size. **Photon-MC tight is the direct comparator for data-tight**: the two agree to ∼ 3% on $\langle n_{\text{owned}}\rangle$ and $\langle \Delta i_\eta\rangle$ across $E_T \in [10, 30]$ GeV. The residual data$\lesssim$photon-MC tendency in $n_{\text{owned}}$ may reflect the BDT cutting more aggressively on data than on MC and is consistent within statistical errors.
- **The $\eta$-asymmetric excess is selection-driven.** Baseline $\langle \Delta i_\eta\rangle$ shows the largest data−MC gap (data 5.65 vs photon MC 3.78 at $E_T = 20$, 1.5 mrad); both cuts collapse it to the MC band (3.63 after tight). $\langle \Delta i_\phi\rangle$ is essentially consistent at every stage (data tight 3.82 vs photon MC tight 3.88) — the excess sits preferentially in $\eta$, consistent with the March-report interpretation.
- **Period differences are small after tight selection.** At $E_T = 20$ GeV, data-tight $\langle n_{\text{owned}}\rangle = 9.92$ (1.5 mrad) vs 9.98 (0 mrad), consistent within statistical errors. Baseline data differs more visibly between periods: 15.86 (1.5 mrad) vs 14.37 (0 mrad), suggesting the 0 mrad broader luminous region pushes more clusters to $|z|$ values where the 7×7 window captures less of the shower. Tight selection absorbs this effect.
- **DI mixing is sub-leading in this selection.** At 1.5 mrad ($f = 7.9\%$) the DI-blend shift in $\langle n_{\text{owned}}\rangle$ is below 0.1 tower for every selection. At 0 mrad ($f = 22.4\%$) the DI blend shifts jet-MC baseline upward by ∼ 0.3 towers (visible in Figs. 4, dashed blue curves) but drops to below 0.1 tower after the tight cut. See Figs. 5, 6 for the pure single vs. double sample-level comparison.

## 1.5 mrad (nominal period)

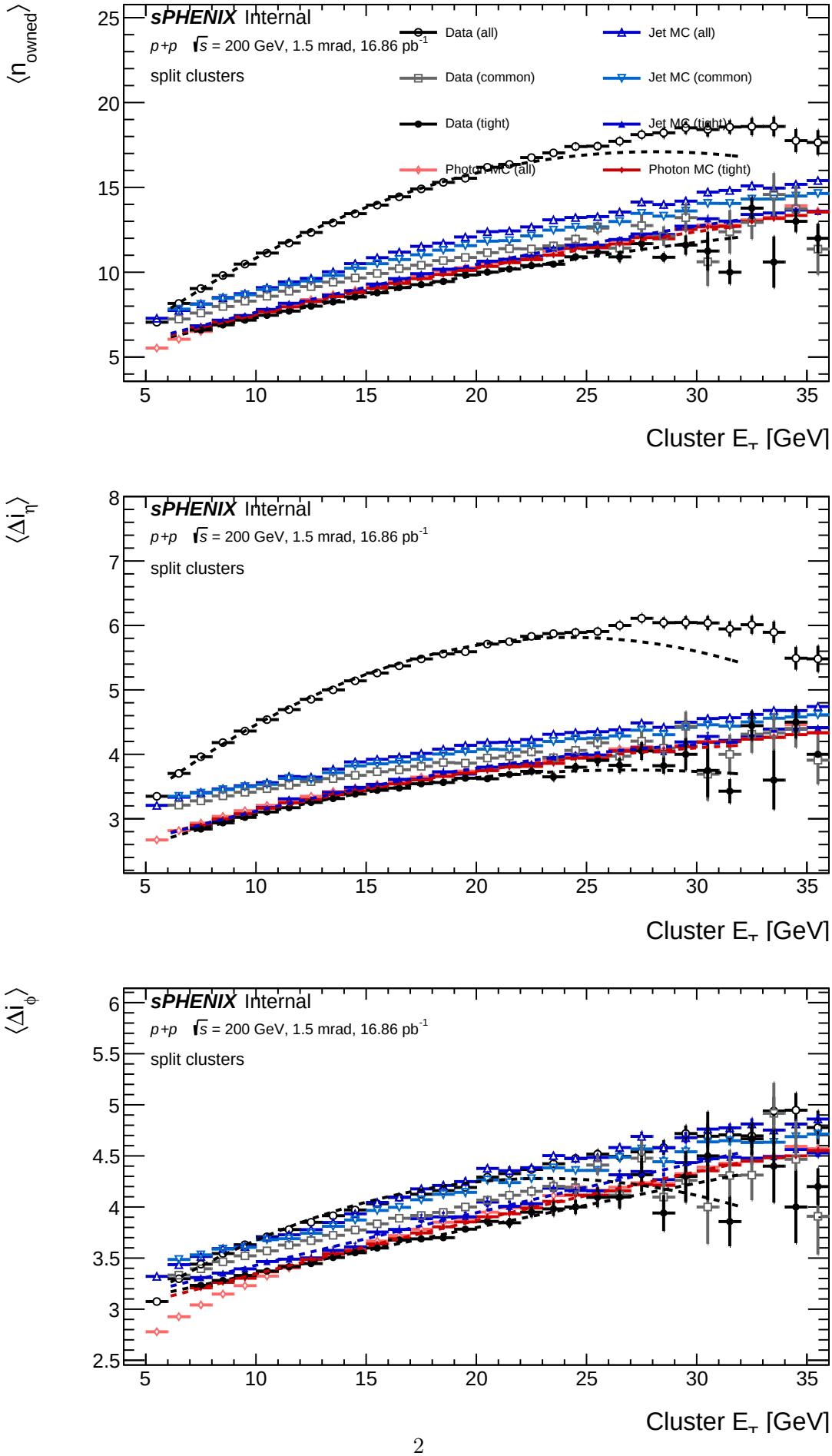




Figure 1: 1.5 mrad, split container. Overlay of data and MC at three selections: baseline (open markers), common-pass (half-emphasized), tight-pass (filled). The photon-MC tight-pass curve is the direct signal-

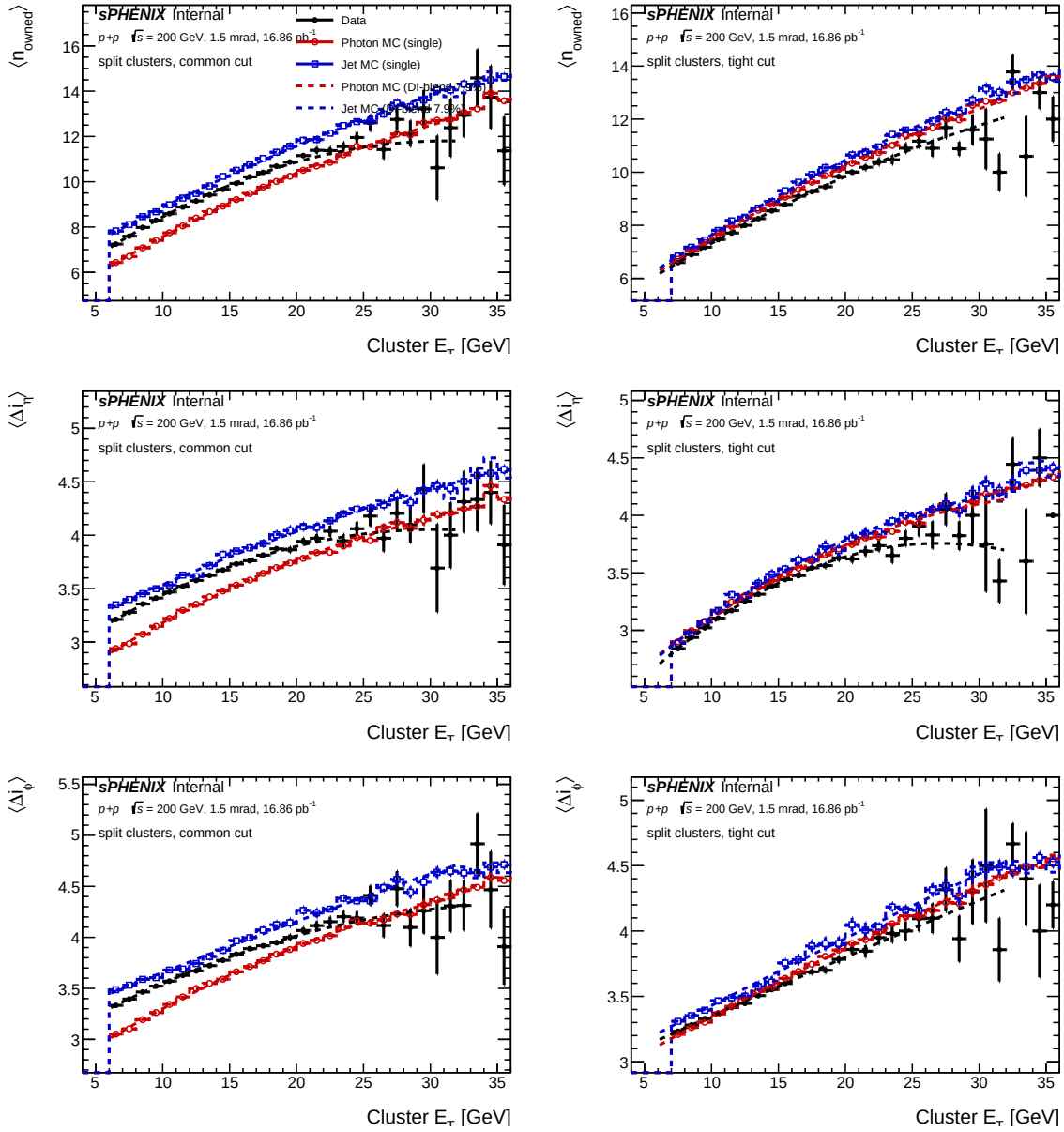


Figure 2: 1.5 mrad, split container. Three metrics (rows) × two selections (columns: common, tight), with the DI-blended MC curves drawn as dashed overlays. Same samples and weighting as Fig. 1. This view makes the MC pooling (single vs DI-blended) easy to read for each selection separately.

## 0 mrad (early period)

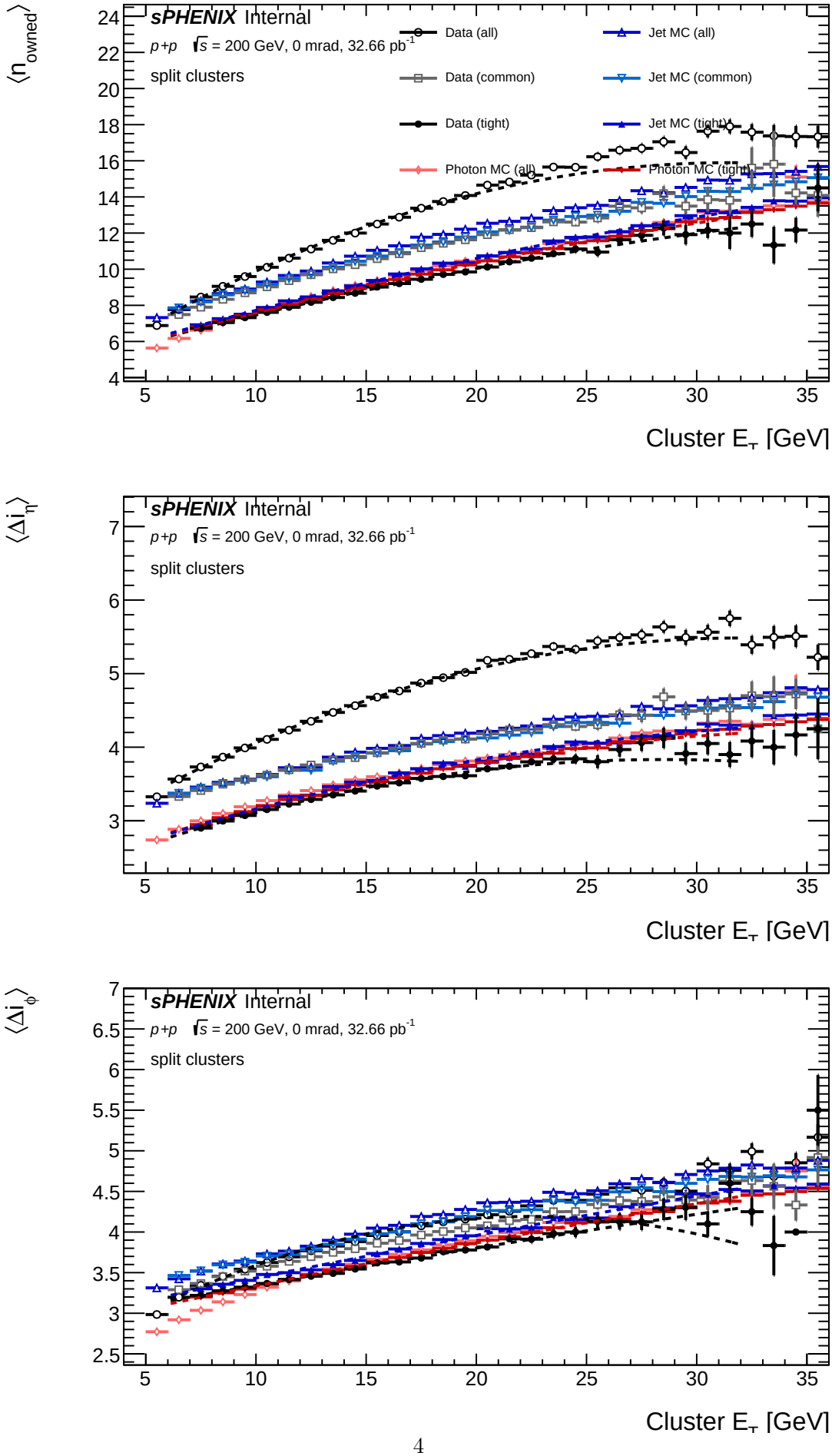




Figure 3: 0 mrad, split container. Same overlay format as Fig. 1. Lumi 32.66 $\mathrm{pb}^{-1}$; truth-vertex reweight is the 0 mrad `reweight.root` (converged in 7 iterations, cf. `truth_vertex_reco_check.pdf`).

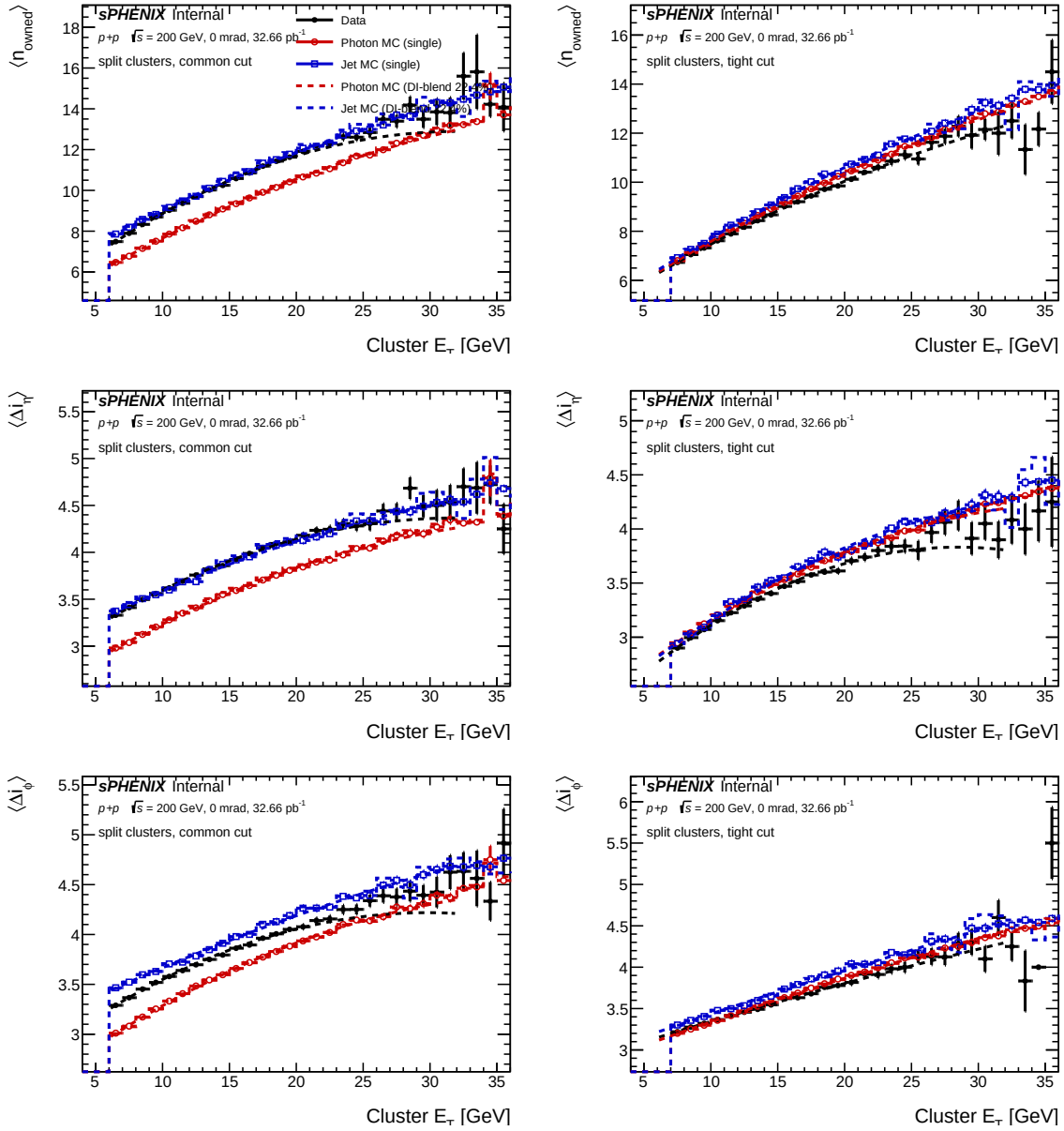


Figure 4: 0 mrad, split container. Same layout as Fig. 2 but for the 0 mrad period.

# Single vs double interaction (pre-cut reference)

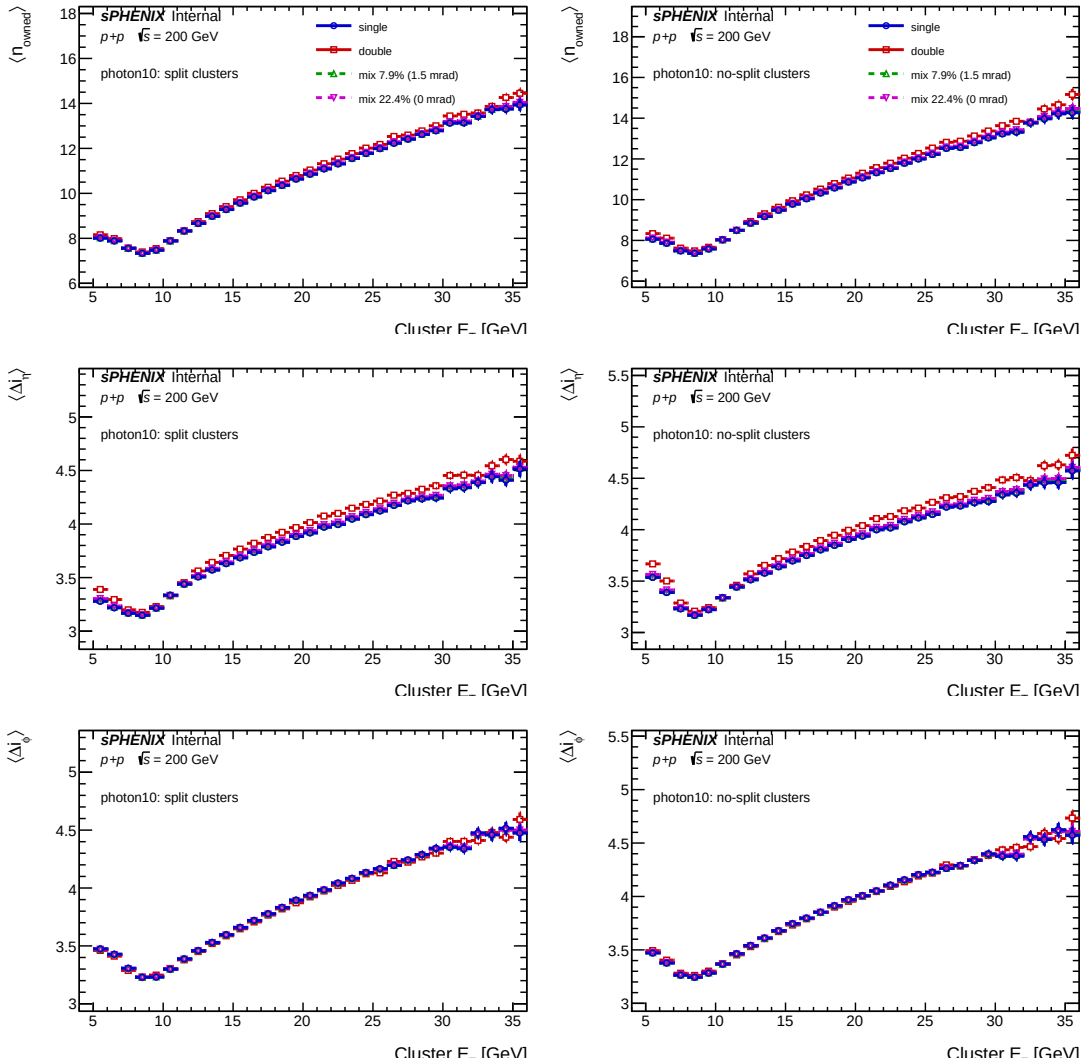


Figure 5: Photon DI reference: `photon10_nom` (blue) vs `photon10_double` (red), plus the two mixed periods ($f = 7.9\%$ green, $f = 22.4\%$ magenta). Pre-cut, both containers. This figure is unchanged from the March short report and does not apply the truth-vertex reweight — it is retained as a direct sample-level reference for the DI effect.

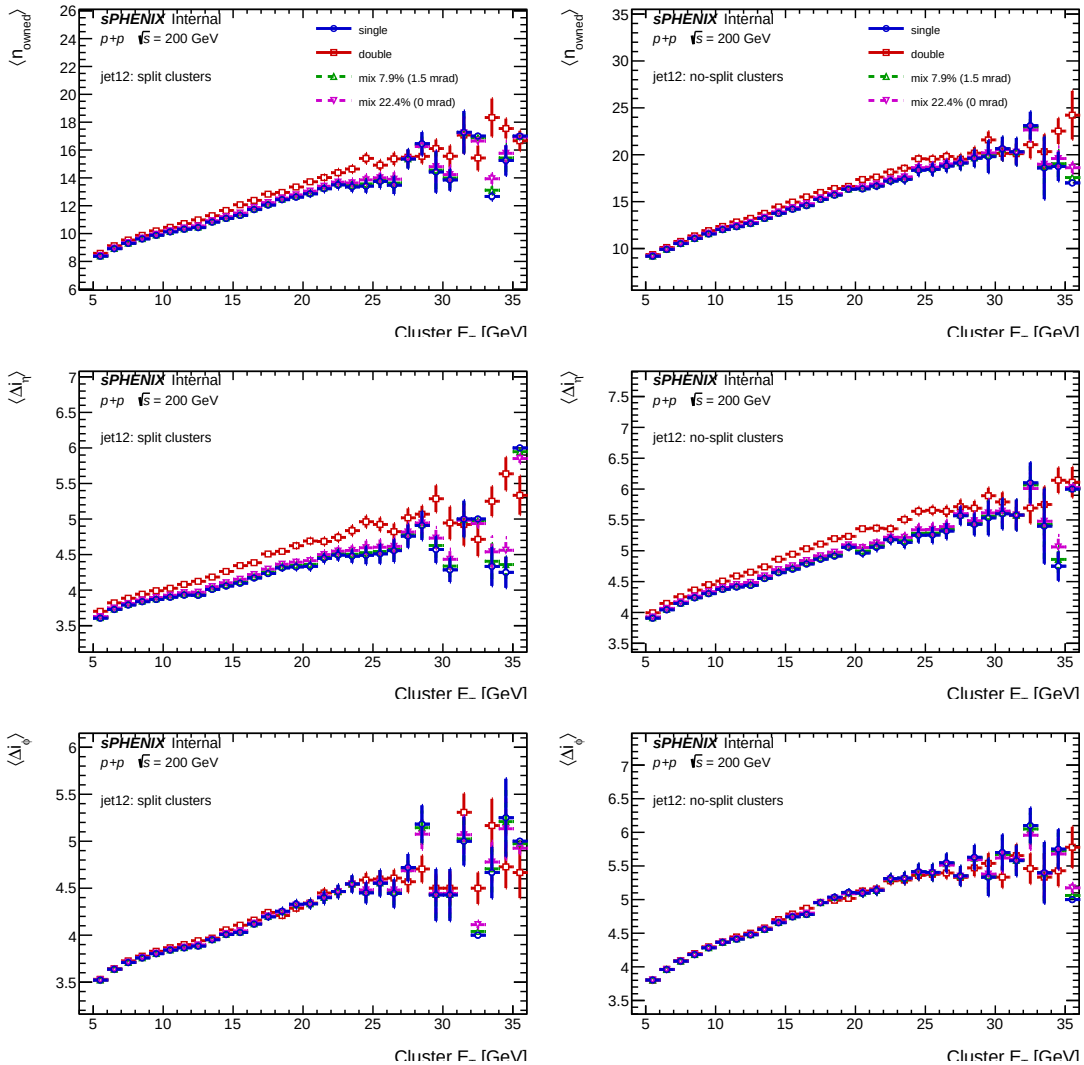


Figure 6: Jet DI reference: `jet12_nom` vs `jet12_double` in the same format as Fig. 5. DI effect on jet cluster size is several times larger than on photon, consistent with softer neighbouring towers getting pushed above the 70 MeV threshold.

## Methods, caveats, reproducibility

**Cuts.** The common and tight blocks are loaded verbatim from the showershape configs. The tight BDT threshold uses the `base_v1E` model (the dominant ET-binned PPG12 choice) with the ET-dependent threshold bdt $> -0.02\,E_T + 0.86$ and the fixed parametric et1, $\langle w_\eta \rangle$, $\langle w_\phi \rangle$ bounds from `tight:` in the YAML. The common block uses prob $\in [0, 1]$, $e_{11}/e_{33} \in [0, 0.98]$, NPB score $> 0.5$, $\langle w_{\eta,\mathrm{cogx}} \rangle < 2$.

**Vertex reweighting.** The nominal method is the data-driven iterative truth-vertex reweight $w(z)$ consumed by `ShowerShapeCheck.C` via `truth_vertex_reweight_on: 1` in the showershape configs and the `h_w_iterative` histogram in `reweight.root`. Per event, the MC weight is multiplied by $w(z_h)$ for single-interaction MC and by $w(z_h)\,w(z_{\mathrm{mb}})$ for double-interaction MC. *Caveat*: the present `photon10_double` and `jet12_double` `bdt_split.root` files were produced before the `vertexz_truth_mb` branch was added to the anatreemaker output, so the secondary-vertex factor is silently dropped for those two samples — an optional-pointer guard identical to the one in `ShowerShapeCheck.C`. The impact on cluster-size means is expected to be below the percent level at 1.5 mrad (where the DI blend fraction is only 7.9%) and sub-percent at 0 mrad (where the MC beam profile is already close to the luminous region). Regenerating these two `bdt_split.root` files from the reprocessed `combined.root` (which does carry `vertexz_truth_mb`) is a follow-up and would remove this caveat.

**DI mixing.** Applied bin-by-bin at the pooled-MC level: $\langle y \rangle_{\mathrm{mix}}(E_T) = \langle y \rangle_{\mathrm{single\ pool}}(E_T) + f\,[\,\langle y \rangle_{\texttt{photon10_double}}(E_T) - \langle y \rangle_{\texttt{photon10}}(E_T)\,]$ (and analogously for the jet pool with `jet12_double`/`jet12`), which avoids biasing the

blend by the per-sample event counts. The pooled single-MC uses unit sample-weights within each class and takes the cross-section-weighted mean via `TProfile::Add`.

**No-split container.** The `photon10_double` and `jet12_double` production predates the dual-container `apply_BDT.C`, so the no-split container lacks BDT and NPB branches for those two samples. The cuts figures for the no-split container therefore *do not* include DI-blended MC; the baseline plots include all samples.

**Reproducibility.**

```
cd efficiencytool
PERIOD=1p5mrad PARALLEL=6 bash run_cluster_size.sh
PERIOD=0mrad  PARALLEL=6 bash run_cluster_size.sh
cd ../plotting
root -l -b -q 'plot_cluster_size_cuts.C+g'
```

# A Per-selection numerical values at $E_T \simeq 20$ GeV

Table values are the mean over bin centres 19.5 and 20.5 GeV of the split-container pooled TProfiles, for each period and each of the three selections.

| | | 1.5 mrad | | | 0 mrad | | |
|---|---|---|---|---|---|---|---|
| metric | sample | baseline | common | tight | baseline | common | tight |
| $\langle n_{\text{owned}} \rangle$ | data | 15.86 | 11.01 | 9.92 | 14.37 | 11.77 | 9.98 |
| | photon | 10.44 | 10.39 | 10.21 | 10.56 | 10.51 | 10.32 |
| | jet | 15.38 | 14.33 | 10.94 | 15.51 | 14.59 | 11.05 |
| $\langle \Delta i_\eta \rangle$ | data | 5.65 | 3.90 | 3.63 | 5.10 | 4.15 | 3.66 |
| | photon | 3.78 | 3.77 | 3.72 | 3.83 | 3.82 | 3.76 |
| | jet | 4.81 | 4.63 | 3.84 | 4.85 | 4.69 | 3.88 |
| $\langle \Delta i_\phi \rangle$ | data | 4.24 | 4.03 | 3.82 | 4.19 | 4.06 | 3.80 |
| | photon | 3.93 | 3.91 | 3.88 | 3.93 | 3.91 | 3.88 |
| | jet | 4.95 | 4.76 | 4.16 | 4.96 | 4.79 | 4.17 |

Table 1: Cluster-size means in the bin centre $E_T \in [19, 21]$ GeV, split container, truth-vertex-reweighted MC. Values read from the per-period TProfiles in `results/cluster_size_cuts_PERIOD_SAMPLE.root` (PERIOD is either 1p5mrad or 0mrad). Photon pool is photon5+photon10+photon20; jet pool is jet5+jet8+jet12+jet20+jet30+jet40. DI blending is applied in the figures only, not in this table, to separate the single-MC pool values from the DI-blend overlay.

**Full narrative report:** `cluster_size_vs_et.pdf` has the sample inventory, detailed per-figure observations, and noise/threshold follow-up discussion that is cross-referenced from the summary bullets above.